

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER

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ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

MR-17 began repeating Test K-44, which was retired 186 days before the incident. K-44 was removed from the active workflow but records do not confirm that every copy, service, or schedule was removed. The operator command to stop MR-17 was validly formatted and returned, "ACTIVE_INSTRUCTIONS_PENDING".

SECTION 02

Verified Facts

- 1. Mara Ilyan (Production Operations Manager) - Said that Nia reported that MR-17 was behaving normally before the incident.
- 2. Maintenance Record: The record does not confirm that every copy, service, or schedule was removed.
- 3. Elias Chen (Robotics Systems Engineer) - The source label is legacy_sync.exe .

SECTION 03

Assumptions and Open Questions

1. Assumption: The maintenance records do not confirm that K-44 has been removed from every possible area MK-17 is in.
2. Open Question: A controlled restart may clear the active state.

SECTION 04**System Environment Map****SECTION 04A****People and Roles**

Nia Voss - Night QA Supervisor
Mara Ilyan - Production
Operations Manager
Elias Chen - Robotics Systems
Engineer

SECTION 04B**Processes and Rules**

Production Operations
Manager controls what work
runs in the QA cell, the
operating schedule, and
software history.

SECTION 04C**Information and Data**

QA lab, Records Terminal, Test
Catalog, Operation Schedule,
Software History

SECTION 04D**Technology and Tools**

MR-17, QA Operations
Interface, The Robot

SECTION 04E**Environment and Constraints**

The only limit is time before the
Robot restarts and clears any
information needed.

SECTION 05

Investigation Boundary

Inside the Investigation

The team needs to investigate the sequence of commands that led to MR-17 repeating retired Test K-44.

Outside for Now

MR-17 repeating retired Test K-44 is a fact and does not need additional investigation.

SECTION 06

Organizational Impact

The QA cell has been isolated so only the current team is affected.

SECTION 07

Problem Statement

There is an informational gap of the software history and maintenance record. The team does not have any info on the software history and maintenance record does not verify that Test K-44 was removed from every aspect and area of the robot.

SECTION 08

Next Investigation Step

The team needs to determine what legacy_sync.exe is and its affect on the robot. That specific unknown file might be the cause of MR-17 breaking out of the scheduled sequence of commands and repeating Test K-44 instead.

SHARED ARTIFACT

Working Document Link

https://github.com/sm333-class/CIT-140-FALL2026-SBVCSM/blob/main/CIT%20140%20Week%201%20Problem%20Brief%20_%20StephaniaMalagon.pdf

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.